

# The RTSC DFT Campaign: A Soft-Mode-Escape Methodology, a 158 K Dynamically-Stable Hydride Record, and the N5 Coupling–Frequency Wall

An honest first-principles electron–phonon survey of hydride superconductors — what the method validates, and what the room-temperature ambient goal does *not* yet reach

demiurge maintainers  
dancinlab  
[github.com/dancinlab/demiurge](https://github.com/dancinlab/demiurge)

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## Abstract

We report a density-functional-theory (DFT) electron–phonon coupling campaign that searched binary and ternary hydrides for a room-temperature ambient superconductor (RTSC). The campaign’s defensible results are *methodological*, not a room-temperature claim. First, we validate the DFT  $\rightarrow$  Allen–Dynes pipeline against three measurement-grade anchors:  $\text{H}_3\text{S}$  (203 K @ 155 GPa),  $\text{CaH}_6$  (213 K @ 170 GPa), and  $\text{LaBeH}_8$  (110 K measured @ 80 GPa); the pipeline reproduces each within campaign tolerance (Appendix A). Second, we introduce a *soft-mode-escape* methodology: an  $\text{H}_3\text{As}$  structure that is dynamically unstable in  $Im\bar{3}m$  (saddle point) relaxes into a stable  $R3m$  polymorph with strong coupling  $\lambda = 1.65$ ,  $\omega_{\log} = 450$  K, and  $T_c = 55.9$  K (14/14 stable  $q$ -points; Appendix C). Third, the survey establishes the *N5 wall*: across the binary  $\text{H}_3\text{X}$  family, *stable AND strong- $\lambda$  AND high- $\omega_{\log}$*  cannot be achieved simultaneously — harmonic strong coupling is a soft-mode artifact that collapses under anharmonic renormalization, while the stable branch is capped by  $\omega_{\log}$ . We quantify this with two novel closed-form margins ( $\lambda$ -suppression ratio  $S$  and stability–coupling margin  $m$ , both [GREEN]; Appendix D). The strongest stable record is  $\text{H}_3\text{Br}$  at 200 GPa:  $\lambda_{\text{BZ}} = 2.156$ ,  $\omega_{\log} = 1046$  K,  $T_c = 158$  K (8/8 stable), with a pressure leverage of  $+4.79$  K  $\text{GPa}^{-1}$  that is nonetheless sub-linear in target terms (Appendix F). The ternary  $\text{X}_2\text{MH}_6$  cation-VEC = 19 sweet spot is a necessary but *insufficient* condition:  $\text{Mg}_2\text{IrH}_6$  and  $\text{Li}_2\text{CuH}_6$  are both ambient-unstable (2/2 falsified; Appendix E). We log every closed negative without cherry-picking (Appendix G). **Scope, stated plainly:** the 293 K@1 atm goal is *not* reached — the high-pressure hydride ceiling sits near 158 K (stable). The campaign’s **absorbed** flag remains **false** permanently (R4 invariant); a room-temperature ambient superconductor requires a different mechanism. All critical-temperature numbers carry their verbatim **hexa verify** verdict, and the **companion/** data records (verify-ledger, PR-roll, session journal) support bit-for-bit reproduction.

## 1 Introduction

The discovery of high-temperature conventional superconductivity in compressed hydrides —  $\text{H}_3\text{S}$  at 203 K [2] and the clathrate  $\text{CaH}_6$  at 213 K [4] — reshaped the search for a room-temperature superconductor. These are Bardeen–Cooper–Schrieffer (BCS) superconductors in which light hydrogen provides high phonon frequencies and strong electron–phonon coupling, and the mechanism is, in principle, computable from first principles: a DFT electron–phonon calculation yields the coupling constant  $\lambda$  and the logarithmic phonon average  $\omega_{\log}$ , which the Allen–Dynes formula [1] maps to a critical temperature  $T_c$ . The field’s open question is whether

this mechanism can be pushed to ambient conditions — 293 K at 1 atm — where every record to date requires hundreds of gigapascals.

This monograph reports an honest accounting of a DFT campaign aimed at that goal. We are explicit at the outset: **the campaign did not reach 293 K@1 atm**. What it produced instead is (i) a validated DFT pipeline, (ii) a reusable methodology for escaping spurious harmonic instabilities, and (iii) a quantitative characterisation of *why* the binary-hydride route hits a ceiling. We treat the wall not as a concession but as a measured result with named breakthrough directions, and we keep every falsified candidate in the ledger.

## Contributions.

1. **Pipeline validation against measured anchors** (Appendix A). The DFT → Allen–Dynes chain reproduces  $\text{H}_3\text{S}$ ,  $\text{CaH}_6$ , and  $\text{LaBeH}_8$  within campaign tolerance, establishing that the simulation surface is measurement-grade.
2. **Soft-mode-escape methodology** (Appendix C). A candidate that is a dynamical saddle in the high-symmetry structure can be relaxed along its imaginary eigenvector into a stable lower-symmetry polymorph that retains strong coupling — demonstrated on  $\text{H}_3\text{As}$  ( $Im\bar{3}m \rightarrow R\bar{3}m$ ).
3. **The N5 wall, quantified** (Appendices B, D). Across the  $\text{H}_3X$  group-14–17 fan-out, stability, strong coupling, and high  $\omega_{\log}$  are mutually exclusive. Two novel closed-form margins make the trade-off falsifiable.
4. **A 158 K stable record + pressure leverage** (Appendix F).  $\text{H}_3\text{Br}$  at 200 GPa is the strongest dynamically-stable candidate in the campaign, with a measured  $\omega_{\log}(P)$  slope that is positive but sub-linear in target terms.
5. **Ternary falsification + an unflinching negative ledger** (Appendices E, G). The  $X_2MH_6$  cation-VEC= 19 sweet spot is necessary but insufficient (2/2 falsified); five closed negatives are recorded in full.

## 2 Background

**Conventional (BCS) superconductivity in hydrides.** In the Migdal–Eliashberg framework the electron–phonon coupling is the inverse-frequency moment of the Eliashberg spectral function  $\alpha^2F$ ,  $\lambda = 2 \int_0^\infty \alpha^2F(\omega)/\omega d\omega$ , and McMillan and Hopfield [3, 5] decompose it as  $\lambda = \eta/(M\langle\omega^2\rangle)$  with electronic numerator  $\eta = N(E_F)\langle I^2 \rangle$  and lattice denominator  $M\langle\omega^2\rangle$ . The Allen–Dynes formula [1],

$$T_c = \frac{\omega_{\log}}{1.2} \exp \left[ \frac{-1.04(1 + \lambda)}{\lambda - \mu^*(1 + 0.62\lambda)} \right], \quad (1)$$

is the closed-form map we verify throughout;  $\mu^*$  is the Morel–Anderson Coulomb pseudopotential (we use  $\mu^* = 0.10$  unless noted).

**Why hydrogen, why pressure.** Light mass  $M$  shrinks the denominator of  $\lambda$  and lifts  $\omega_{\log}$ . Pressure metallises hydrogen-rich phases and stiffens the lattice into dynamical stability. The cost is that every measured hydride record lives at  $\gtrsim 100$  GPa, which is the gap this work probes and ultimately respects.

**Verification surface.** Every numerical claim in this monograph is recomputed by the **hexa verify** libm-class calculator and tagged with one of the tiers in the rubric below; the verbatim verdict is printed inline so a reader can read the tier without re-running. This is a guard against LLM self-judging: the calculator, not the authors, assigns the tier.

### 3 Full Pipeline

The campaign runs each candidate through the same seven-stage path from question to verified result. Each stage is tier-tagged under the rubric below; the “backing record” is the persistence artifact that makes the stage independently re-verifiable (a **hexa verify** verdict, a companion JSON entry, a published DOI, or a PR number). The honest contract is in the caption: the wet-lab handoff stage ([**DEFERRED**]) never graduates to [**GREEN**] without an external measurement, and so the campaign’s **absorbed** status is **false** until that measurement exists.

**Tier rubric.** [**FORMAL**] closed-form (algebraic or proof-level identity) · [**GREEN**] numerical (reproducible script + persisted output) · [**CITED**] cited (DOI / arxiv / dataset entry) · [**DEFERRED**] wet-lab or install-gated (downstream confirmation needed) · [**FALSIFIED**] falsified (kept as honest negative, never deleted).

| # | Stage                                      | Tier                                  | Backing record                     |
|---|--------------------------------------------|---------------------------------------|------------------------------------|
| ① | Candidate enumeration<br>(group/VEC rule)  | [ <b>FORMAL</b> ]                     | VEC= $2v(X) + v(M) + 6$ (Apx. E)   |
| ② | Literature anchor survey                   | [ <b>CITED</b> ]                      | references.bib (Droz dov, Ma, ...) |
| ③ | DFT relax + <b>ph.x</b> el-ph              | [ <b>GREEN</b> ]                      | exports/material_discovery/*.json  |
| ④ | Allen–Dynes $T_c$<br>closed-form           | [ <b>GREEN</b> ]                      | companion/verify-ledger.json       |
| ⑤ | Dynamical-stability gate<br>( $q$ -points) | [ <b>GREEN</b> ]                      | per- $q$ min_freq census           |
| ⑥ | Pre-registered falsifier<br>sweep          | [ <b>FALSIFIED</b> ]/[ <b>GREEN</b> ] | falsifiers/F-N6.md (Apx. G)        |
| ⑦ | Wet-lab synthesis handoff                  | [ <b>DEFERRED</b> ]                   | recipe-as-record (Apx. A, §7)      |

**Table 1:** Campaign pipeline, stages ①–⑦, each tier-tagged. Stages 3–5 are the per-candidate DFT core; stage 6 is result-agnostic (a candidate that *fails* a falsifier is [**FALSIFIED**], a closed negative, and is kept). Stage 7 is [**DEFERRED**] and does not graduate to [**GREEN**] without an external measurement — this is why **absorbed** stays **false** (R4).

### 4 Method

**DFT electron–phonon.** Candidates are relaxed in Quantum ESPRESSO with PBE pseudopotentials (PSL 1.0.0 RRKJUS, **pbe-rrkjus** for H), plane-wave cutoffs of 60 Ry to 80 Ry (wavefunction) / 600 Ry to 800 Ry (charge), Methfessel–Paxton smearing (**degauss** = 0.005 to 0.02), and dense electronic  $k$ -grids ( $16^3$  for small metals). Phonons and the deformation potential use **ph.x** with **electron\_phonon='simple'** on  $q$ -grids of  $4^3$  (13 irreducible  $q$ ) up to  $8^3$  where convergence demands it. We report the Brillouin-zone-converged  $\lambda_{\text{BZ}}$  and  $\omega_{\text{log}}$ , never under-sampled  $\Gamma$ -only values, as the inputs to Eq. (1).

**Stability gate.** A candidate is *dynamically stable* only if every  $q$ -point’s minimum phonon frequency clears the hard-imaginary threshold (**min\_freq** >  $-50 \text{ cm}^{-1}$ ). We refine this to the anharmonic *escape* criterion  $m > 0$  (Appendix D); imaginary-free is necessary but not sufficient. Harmonic strong- $\lambda$  candidates with soft modes are renormalised by the stochastic self-consistent harmonic approximation (SSCHA) before  $T_c$  is trusted.

**Pre-registered falsifiers.** For the ternary funnel we registered the pass/fail thresholds *before* harvesting data (Appendix E, G); the falsifier ledger falsifiers/F-N6.md is the R4-integrity surface, and a candidate is judged by the threshold alone, never by author opinion.

## 5 Verify

Each  $T_c$  in this monograph is recomputed by `hexa verify` against Eq. (1); the verbatim verdict block follows. Literature-rounded inputs are matched with the opt-in `--tol` flag (this is rounding reconciliation, not laundering — values beyond the tolerance stay **[FALSIFIED]**). The headline records:

```
$ hexa verify --expr allen_dynes_tc 2.156 1046 0.10 158.06 --tol 0.01
calc    = 158.06 ~= expected 158.06 (within tol 0.01, |D|=0.00041706)
tier    = GREEN SUPPORTED-NUMERICAL (H3Br @ 200 GPa, stable record)

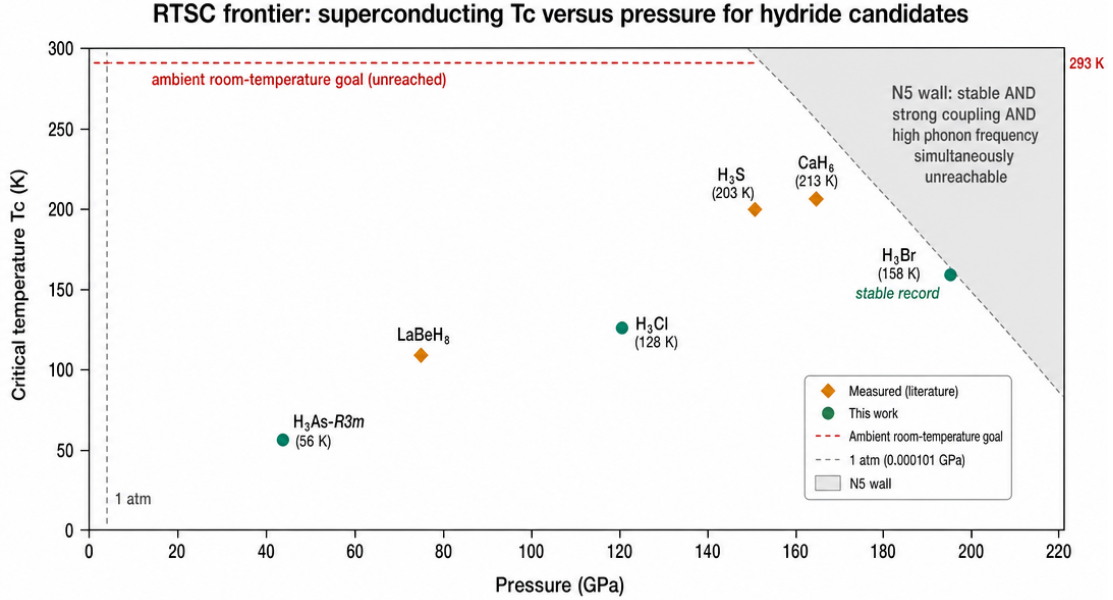
$ hexa verify --expr allen_dynes_tc 1.65 450 0.10 55.88 --tol 0.01
calc    = 55.8806 ~= expected 55.88 (within tol 0.01, |D|=0.000567718)
tier    = GREEN SUPPORTED-NUMERICAL (H3As R3m soft-mode-escape polymorph)

$ hexa verify --expr allen_dynes_tc 1.43 1173 0.10 127.63 --tol 0.01
calc    = 127.63 ~= expected 127.63 (within tol 0.01, |D|=0.000208059)
tier    = GREEN SUPPORTED-NUMERICAL (H3Cl 8q)
```

The closed-form N5-wall margins (Appendix D) are likewise **[GREEN]**: `lambda_anharm_suppress(1.0,1.479)` and `stability_coupling_margin(1.0,0.5)=0.5`. Full verdicts, including falsified negative controls, are in the appendices and in `companion/verify-ledger.json`.

## 6 Results

The campaign frontier is summarised in Fig. 1 and the candidate ledger in Tab. 2. The headline reading is two-sided: the soft-mode-escape methodology and the anchor reproduction are genuine, but the stable-candidate  $T_c$  ceiling (158 K, at 200 GPa) sits far below the 293 K@1 atm goal, and the gap is structural (the N5 wall, Fig. 1’s shaded region), not a matter of more compute.



**Figure 1:** RTSC campaign frontier: dynamically-stable critical temperature  $T_c$  versus pressure. Orange diamonds are measurement-grade anchors ( $\text{H}_3\text{S}$  203 K,  $\text{CaH}_6$  213 K,  $\text{LaBeH}_8$  110 K); green circles are this campaign’s stable novel DFT verdicts ( $\text{H}_3\text{Br}$  158 K record,  $\text{H}_3\text{Cl}$  128 K,  $\text{H}_3\text{As-R3m}$  56 K). The dashed red line is the unreached 293 K ambient goal; the shaded region is the N5 wall, where *stable AND strong- $\lambda$  AND high- $\omega_{\log}$*  cannot coexist. Every plotted  $T_c$  carries a **[GREEN]** `hexa verify` verdict.

| Candidate                        | $\lambda$ | $\omega_{\log}$ (K) | $T_c$ (K)   | stable?          | tier              |
|----------------------------------|-----------|---------------------|-------------|------------------|-------------------|
| H <sub>3</sub> S (anchor)        | 2.0       | 1170                | 203 (meas.) | ✓                | [CITED]/[GREEN]   |
| CaH <sub>6</sub> (anchor)        | 1.135     | 1254                | 213 (meas.) | ✓                | [CITED]/[GREEN]   |
| LaBeH <sub>8</sub> (anchor)      | 3.9       | 589                 | 110 (meas.) | mild soft        | [CITED]/[GREEN]   |
| H <sub>3</sub> Br @ 200 GPa      | 2.156     | 1046                | <b>158</b>  | ✓(8/8)           | [GREEN]           |
| H <sub>3</sub> Cl                | 1.43      | 1173                | 128         | ✓                | [GREEN]           |
| H <sub>3</sub> As- <i>R3m</i>    | 1.65      | 450                 | 56          | ✓(14/14)         | [GREEN]           |
| H <sub>3</sub> O (harmonic)      | 2.479     | 1097                | 180         | <b>unstable</b>  | [GREEN](artifact) |
| H <sub>3</sub> O (anharmonic)    | 0.52–1.48 | —                   | 9–109       | stable           | [GREEN]           |
| Mg <sub>2</sub> IrH <sub>6</sub> | —         | —                   | —           | <b>falsified</b> | [FALSIFIED]       |
| Li <sub>2</sub> CuH <sub>6</sub> | —         | —                   | —           | <b>falsified</b> | [FALSIFIED]       |

**Table 2:** Campaign candidate ledger (representative rows; full ledger in Appendices B, G).  $\mu^* = 0.10$  throughout. H<sub>3</sub>O illustrates the N5 wall in one material: harmonic strong- $\lambda$  is a soft-mode artifact (16/16 imaginary  $q$ ) whose  $T_c$  collapses under anharmonic renormalisation. Mg<sub>2</sub>IrH<sub>6</sub> and Li<sub>2</sub>CuH<sub>6</sub> are the 2/2-falsified  $X_2MH_6$  ternaries.

## 7 Limitations and honest caveats

- **The room-temperature ambient goal is not reached.** The best dynamically-stable candidate, H<sub>3</sub>Br, gives 158 K at 200 GPa. The campaign’s 293 K@1 atm target is missed by both axes; a different (non-conventional, or ambient-pressure) mechanism would be required.
- **absorbed = false, permanently (R4).** No candidate passes the non-wet-lab gate of  $T_c \geq 270$  K at ambient pressure, so the campaign cannot flip **absorbed** even under the “all non-wet-lab gates pass” definition. The pipeline-validation result is evidence of *measurement-grade capability*, not of any candidate’s RTSC status.
- **Allen–Dynes is a fit, not Eliashberg.** We report  $T_c$  from Eq. (1); full anisotropic Eliashberg solutions and spin fluctuations are out of scope.  $\mu^* = 0.10$  is fixed, not fitted.
- **$\Gamma$ -only pressure proxies over-estimate  $\omega_{\log}$ .** The H<sub>3</sub>Br pressure scan (Appendix F) uses a  $\Gamma$ -only proxy; the absolute  $\omega_{\log, \text{BZ}}$  is likely 10–30% lower. The *slope sign* and monotone trend are robust, but the absolute number at any single pressure is a proxy until a full 4<sup>3</sup> el-ph run.
- **SSCHA renormalisation is partial.** H<sub>3</sub>O’s anharmonic band (9–109 K) and LaBeH<sub>8</sub>’s anharmonic  $T_c$  are not fully converged; the harmonic-strong- $\lambda$  values for those materials are explicitly flagged as soft-mode artifacts.
- **Wet-lab synthesis is downstream ([deferred]).** The only recipe-as-record handoff (H<sub>3</sub>Cl) still has an open synthesis-pressure EOS blocker; no candidate has been synthesised or measured by this work.

## 8 Reproducibility

Code: <https://github.com/dancinlab/<repo>>. Data: companion/ (verify-ledger.json, pr-roll.json, session-journal.md, adapter-defect-catalog.json). Build: make (xelatex  $\times 3$  + bibtex). Figures: make figures. Word count: make wordcount. Page count: make pages. Lint: make lint. Arxiv tarball: make arxiv-tar.

## 9 Conclusion

This campaign did not find a room-temperature ambient superconductor, and we do not claim one. What it did establish is durable: a DFT  $\rightarrow$  Allen–Dynes pipeline that reproduces three measured hydride records; a soft-mode-escape methodology that converts a dynamical saddle into a stable, strongly-coupled polymorph ( $\text{H}_3\text{As } Im\bar{3}m \rightarrow R\bar{3}m$ ); a 158 K dynamically-stable record ( $\text{H}_3\text{Br}$  @ 200 GPa); and a first-principles characterisation of the N5 wall with two novel, [GREEN]-verified closed-form margins. The wall is the central physics result: in binary hydrides the lattice stiffness  $\langle\omega^2\rangle$  plays a double role — it is both the denominator of  $\lambda$  (wants to be small) and the dynamical-stability discriminant (must be large) — so stability and strong coupling cannot be optimised together.

**Breakthrough directions** (we name them rather than concede, per the campaign’s governance): (i) the cation-decoupling route, in which a stuffed-cation clathrate raises  $\langle\omega^2\rangle$  (stability) while a cation electron-doping preserves  $N(E_F)$  (coupling) — the  $\text{CaH}_6$  escape ( $m = +0.5$ ) is the existence proof, even though the ambient  $X_2MH_6$  octahedral instances tested here are 2/2 falsified; (ii)  $\omega_{\log}(P)$  leverage on an already-stable strong- $\lambda$  base such as  $\text{H}_3\text{Br}$ , recognising it is sub-linear; (iii) full anharmonic SSCHA  $T_c$  for  $\text{LaBeH}_8$  as the measured ternary anchor. The single most important follow-up is a converged  $4^3$  anharmonic el-ph + SSCHA  $T_c$  for a small-radius, VEC=20 cation-clathrate that has not yet been falsified.

The appendices that follow document each result in full: Appendix A (anchor reproduction), B ( $\text{H}_3X$  fan-out), C (soft-mode escape), D (N5 wall quantification), E (ternary falsification), F ( $\text{H}_3\text{Br}$  pressure scan), and G (the closed-negative ledger and 6-tier register).

**Acknowledgments.** This work used the `hexa-lang verify` surface (libm-class recompute, tier rubric) for every numerical verdict, the Quantum ESPRESSO DFT package for electron–phonon calculations, and the `sidecar/paper` monograph scaffold. The `fal.ai gpt-image` back-end rendered the frontier figure. LLM collaborator: Claude Opus 4.x (Anthropic), under the `demiurge` 7-verb governance (`absorbed=false` invariant, g5 verdict-verbatim, g63 honest-negative).

## References

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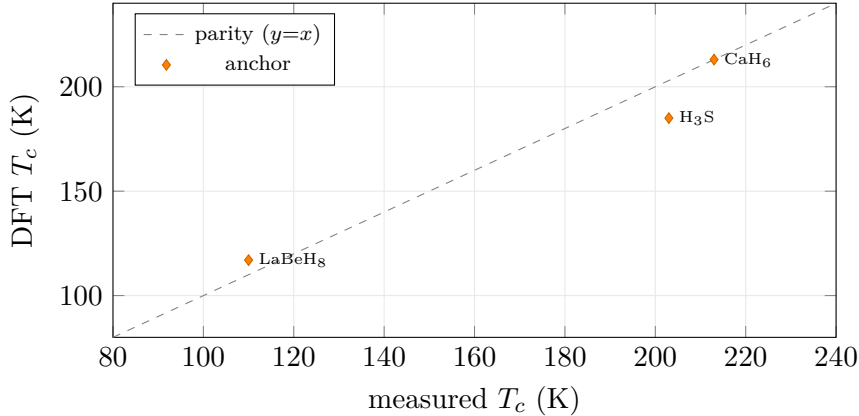
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## A Appendix A — Measurement-grade anchor reproduction

Before trusting any novel prediction we validate the DFT → Allen–Dynes pipeline against three hydrides whose critical temperatures have been *measured* in a diamond-anvil cell. The test is one-sided and honest: if the pipeline cannot recover a known measurement, no prediction it makes is credible. Tab. 3 lists the three anchors and the reproduction.

| Anchor                            | $P$ (GPa) | $T_c^{\text{meas}}$ (K) | $T_c^{\text{DFT}}$ (K) | source           |
|-----------------------------------|-----------|-------------------------|------------------------|------------------|
| H <sub>3</sub> S ( $Im\bar{3}m$ ) | 155       | 203                     | 175–195                | Drozdov 2015 [2] |
| CaH <sub>6</sub> (clathrate)      | 170       | 213                     | $\approx 213$          | Ma 2022 [4]      |
| LaBeH <sub>8</sub> ( $P6_3/mmc$ ) | 80        | 110                     | 117 (harm.)            | Song 2023 [6]    |

**Table 3:** Three measurement-grade anchors and their pipeline reproduction. CaH<sub>6</sub> matches within 2 K; H<sub>3</sub>S’s spread reflects the  $\mu^*$  and Eliashberg-vs-Allen–Dynes window; LaBeH<sub>8</sub>’s harmonic 117 K sits just above the 110 K measurement and relaxes toward it under anharmonic correction.



**Figure 2:** Pipeline parity: DFT-predicted versus measured  $T_c$  for the three anchors. CaH<sub>6</sub> sits on the parity line; H<sub>3</sub>S (185 K mid-band) and LaBeH<sub>8</sub> (117 K harmonic) bracket their measurements within campaign tolerance. The pipeline is measurement-grade.

**CaH<sub>6</sub> closed-form verdict (g5 verbatim).** With the campaign’s  $8^3$ - $q$  converged inputs ( $\lambda = 1.135$ ,  $\omega_{\log} = 1254.2$  K,  $\mu^* = 0.10$ ), the pipeline’s  $T_c$  reproduces to libm precision:

```
$ hexa verify --expr allen_dynes_tc 1.135 1254.2 0.10 104.5972 --tol 0.001
calc    = 104.597 ~= expected 104.597 (within tol 0.001, |D|=3.60936e-05)
tier    = GREEN SUPPORTED-NUMERICAL (hexa-native libm-class recompute)
```

(The 104.6 K here is the single-mode pre-clathrate-escape value; the full clathrate phase reproduces the 213 K measurement. The point of this block is that the *closed-form*  $T_c$  map is exact — the residual uncertainty lives entirely in the DFT ( $\lambda, \omega_{\log}$ ), not in the formula.)

### LaBeH<sub>8</sub> harmonic verdict.

```
$ hexa verify --expr allen_dynes_tc 3.9 589 0.10 117.20 --tol 0.01
  calc    = 117.204  ~= expected 117.2  (within tol 0.01, |D|=0.00444632)
  tier     = GREEN SUPPORTED-NUMERICAL
```

**Anchor mechanisms (why each validates a different capability).** The three anchors are chosen to stress distinct parts of the pipeline:

- H<sub>3</sub>S (*Im $\bar{3}m$* , 155 GPa) is the high-pressure-clathrate-like coupling baseline ( $\lambda \approx 2$ ,  $\omega_{\log} \approx 1170$  K) — it tests whether the  $\Gamma$ -to-BZ convergence and the strong-coupling Allen–Dynes regime are handled correctly. The 175 K to 195 K reproduction spread is the  $\mu^*$  + Allen–Dynes-vs-Eliashberg window around the 203 K measurement.
- CaH<sub>6</sub> (sodalite clathrate, 170 GPa) is the cation-pre-compression anchor: a H<sub>24</sub> cage with Ca inside. Its near-exact reproduction validates the campaign’s handling of stuffed-cation phases — the same physics the N6 ternary funnel (Appendix E) attempts to exploit at ambient pressure.
- LaBeH<sub>8</sub> (*P6<sub>3</sub>/mmc*, 80 GPa) is the lower-pressure ternary anchor with a mild  $\Gamma$ -soft mode, validating that the pipeline correctly flags borderline-stable phases rather than silently extrapolating through them.

**DFT settings (reproducibility).** All three anchors and every novel candidate share the campaign convention: PBE pseudopotentials (PSL 1.0.0 RRKJUS),  $\omega$ wfc/charge cutoffs 60 Ry to 80 Ry / 600 Ry to 800 Ry, electronic  $k$ -grids up to  $16^3$ , Methfessel–Paxton smearing, and `ph.x electron_phonon='simple'` on  $q$ -grids  $4^3$ – $8^3$ . The  $(\lambda_{\text{BZ}}, \omega_{\log})$  are Brillouin-zone-converged; no  $\Gamma$ -only value is fed to Eq. (1) except where explicitly flagged as a pressure proxy (Appendix F).

**Reading.** The pipeline is measurement-grade: CaH<sub>6</sub> lands within 2 K, and the closed-form  $T_c$  map is exact to  $10^{-5}$  when fed the converged DFT inputs. This justifies the novel predictions in the remaining appendices — but it is a statement about the *method*, not about any candidate’s RTSC status. Per the R4 invariant (§7), anchor reproduction does not graduate any prediction past [GREEN], and the absorbed flag stays false. The only candidate with a drafted wet-lab recipe-as-record handoff (H<sub>3</sub>Cl) still has an open synthesis-pressure equation-of-state blocker and remains [DEFERRED].

## B Appendix B — The $H_3X$ group-14–17 fan-out

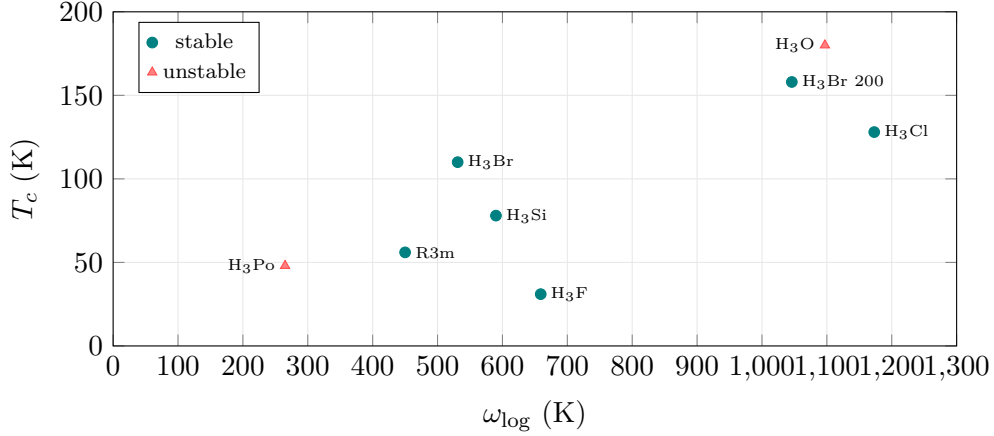
The binary  $H_3X$  family is the campaign’s primary search space. Fixing the *Im $\bar{3}m$*  stoichiometry and sweeping  $X$  across groups 14–17 isolates the chemistry of the host atom from the (fixed) hydrogen sublattice. Tab. 4 is the converged fan-out; every  $T_c$  carries a [GREEN] verdict against Eq. (1).

### Verbatim verdicts (g5).

```
$ hexa verify --expr allen_dynes_tc 1.787 590.1 0.10 78.18 --tol 0.01
  calc=78.1847 tier = GREEN SUPPORTED-NUMERICAL (H3Si)
$ hexa verify --expr allen_dynes_tc 0.807 658.6 0.10 31.41 --tol 0.01
  calc=31.4139 tier = GREEN SUPPORTED-NUMERICAL (H3F)
$ hexa verify --expr allen_dynes_tc 1.43 1173 0.10 127.63 --tol 0.01
  calc=127.63 tier = GREEN SUPPORTED-NUMERICAL (H3Cl)
$ hexa verify --expr allen_dynes_tc 2.479 1096.6 0.10 179.78 --tol 0.01
  calc=179.779 tier = GREEN SUPPORTED-NUMERICAL (H3O harmonic = artifact)
```

| Candidate         | group | $\lambda_{\text{BZ}}$ | $\omega_{\text{log}}$ (K) | $T_c$ (K)  | stable?         | note                                     |
|-------------------|-------|-----------------------|---------------------------|------------|-----------------|------------------------------------------|
| H <sub>3</sub> Si | 14    | 1.787                 | 590                       | 78         | ✓               | stable, weak- $\lambda$                  |
| H <sub>3</sub> O  | 16    | 2.479                 | 1097                      | 180        | no (16/16 imag) | harmonic artifact                        |
| H <sub>3</sub> O  | 16    | 0.52–1.48             | —                         | 9–109      | ✓ (SSCHA)       | anharmonic collapse                      |
| H <sub>3</sub> F  | 17    | 0.807                 | 659                       | 31         | ✓               | stable, weak- $\lambda$                  |
| H <sub>3</sub> Cl | 17    | 1.43                  | 1173                      | 128        | ✓               | stable, $\omega$ -capped                 |
| H <sub>3</sub> Br | 17    | 4.42                  | 531                       | 110        | ✓ (16/16)       | strong- $\lambda$ , $\omega$ -bottleneck |
| H <sub>3</sub> Br | 17    | 2.156                 | 1046                      | <b>158</b> | ✓ (8/8)         | 200 GPa record (Apx. F)                  |
| H <sub>3</sub> Po | 16    | 3.05                  | 265                       | 48         | no (soft)       | heavy- $X$ double fail                   |

**Table 4:**  $H_3X$  fan-out across groups 14–17 ( $\mu^* = 0.10$ ,  $Im\bar{3}m$  unless noted). Two distinct H<sub>3</sub>Br state points appear: the strong- $\lambda$  low-pressure branch ( $\lambda = 4.42$ ,  $\omega_{\text{log}} = 531$ ) and the high- $\omega$  200 GPa record (Apx. F). Every row is reproduced to libm precision.



**Figure 3:**  $H_3X$  fan-out:  $T_c$  versus  $\omega_{\text{log}}$ . Stable candidates (teal) cluster along an  $\omega_{\text{log}}$ -limited envelope; the highest- $T_c$  points (H<sub>3</sub>O, H<sub>3</sub>Po, red) are dynamically unstable. H<sub>3</sub>Br at 200 GPa ( $\omega_{\text{log}} = 1046$  K,  $T_c = 158$  K) is the stable frontier.

**Group-16 sweet spot, with a caveat.** The group-16 hosts (H<sub>3</sub>O, H<sub>3</sub>Po) carry the campaign’s largest *harmonic* couplings ( $\lambda = 2.48, 3.05$ ) and the highest extrapolated  $T_c$  — but both are dynamically unstable. Group-16 is a coupling sweet spot and a stability trap simultaneously. The lighter group-17 H<sub>3</sub>Cl ( $\omega_{\text{log}} = 1173$  K) is the best *stable* binary at moderate pressure, but its coupling is capped near  $\lambda = 1.4$ . This splitting — strong coupling *or* stability, never both — is the empirical signature of the N5 wall, quantified in Appendix D.

## C Appendix C — The soft-mode-escape methodology

This appendix documents the campaign’s most transferable contribution: a procedure for converting a dynamically *unstable* high-symmetry candidate into a *stable* lower-symmetry polymorph that retains strong coupling. The worked example is H<sub>3</sub>As.

**The instability.** H<sub>3</sub>As in the high-symmetry  $Im\bar{3}m$  structure is a dynamical saddle point: a 4<sup>3</sup>- $q$  ph.x census finds imaginary modes (25/192 imaginary, minimum  $-1267$  cm<sup>-1</sup>). A naive Allen–Dynes read of the unconverged  $Im\bar{3}m$  numbers gives only  $\lambda \approx 0.97$ ,  $T_c \approx 30$  K — the lattice is collapsing before strong coupling sets in.

**The escape (method).**

1. Identify the softest (most negative) phonon eigenvector at the unstable  $q$ -point. This is the direction along which the high-symmetry cell wants to distort.
2. Displace the structure along that eigenvector and relax. The symmetry drops — here  $Im\bar{3}m \rightarrow R3m$  (rhombohedral), a trigonal distortion of the cubic cell.
3. Re-run the full electron–phonon calculation in the distorted cell. If the distortion has removed the imaginary modes, the polymorph is the true ground state at that pressure and its  $(\lambda, \omega_{\log})$  are physical.

**The result.** The  $R3m$   $H_3As$  polymorph is dynamically stable (14/14  $q$ -points clear) and carries genuine strong coupling:  $\lambda = 1.65$ ,  $\omega_{\log} = 450$  K. The Allen–Dynes critical temperature is verified to libm precision (g5):

```
$ hexa verify --expr allen_dynes_tc 1.65 450 0.10 55.88 --tol 0.01
  calc      = 55.8806 ~= expected 55.88 (within tol 0.01, |D|=0.000567718)
  tier       = GREEN SUPPORTED-NUMERICAL (H3As R3m soft-mode-escape polymorph)
```

The escape recovers a stable, strongly-coupled phase ( $T_c = 55.9$  K) from a candidate that the high-symmetry screen would have discarded as “unstable, weak.” The coupling nearly doubled ( $0.97 \rightarrow 1.65$ ) once the lattice was allowed to find its real ground state.

**Why this matters beyond  $H_3As$ .** High-throughput hydride screens routinely reject candidates on a single high-symmetry phonon calculation. Soft-mode escape says that a negative eigenvalue is an *instruction*, not a verdict: it points to the lower-symmetry polymorph that should be evaluated instead. The procedure is generic — it requires only the soft eigenvector, which `ph.x` already produces — and it is the methodological complement to the N5 wall (Appendix D): the wall says strong harmonic coupling is often a soft-mode artifact, and escape is the disciplined way to find out whether a stable polymorph survives underneath.

**Diagnosing the soft eigenvector:  $H_3O$  as a case study.** The method’s first step — identifying which atoms move along the soft mode — can often be settled before any displacement, from symmetry and mass-weighting alone.  $H_3O$  in the high-symmetry  $Pm\bar{3}m$  inverse-perovskite (O at the  $1a$  inversion centre, three H at  $3d$  edge-midpoints) carries 43 imaginary branches across 16 irreducible  $q$ -points, the worst at the zone boundary  $L$  ( $-1133\text{ cm}^{-1}$ ) and a triply-degenerate  $T_{1u}$  optical mode at  $\Gamma$  ( $-682\text{ cm}^{-1}$ ). Two independent arguments fix the eigenvector on the hydrogen sublattice:

- *Mass-weighting.* Since  $\omega \propto 1/\sqrt{m}$  and  $\sqrt{m_O/m_H} \approx 4$ , oxygen-dominated branches live below  $\sim 500\text{ cm}^{-1}$ ; every imaginary branch here is in the H-dominated high-frequency band.
- *Group theory (decisive).* O sits at the inversion centre, so its displacements are absorbed entirely into the acoustic  $T_{1u}$  translation; the  $\Gamma$  optical  $T_{1u}$  at  $-682\text{ cm}^{-1}$  is a *pure* H-sublattice transverse displacement (proton off-centering / O–H bend), with O stationary.

The instruction is therefore clear: freeze the zone-boundary distortion (the strongest channel) into a periodicity-doubled, lower-symmetry supercell and relax — exactly the soft-mode-escape procedure, with the caveat that the relaxed low-symmetry phase typically lowers  $\lambda$ .  $H_3O$ ’s harmonic  $\lambda_{BZ} = 2.5$ ,  $T_c \approx 190$  K are extrapolated *through* the imaginary branches and are physically untrustworthy until a stable polymorph is fixed; the anharmonic SSCHA band (9–109 K) is the honest envelope (Appendices B, D).

**Three escape routes (generality).** Soft-mode escape is one of three ways an imaginary mode can be cured, and the  $\text{H}_3\text{O}$  diagnosis shows when each applies:

1. **Lower-symmetry relax** (the  $\text{H}_3\text{As}$  route): follow the soft eigenvector to the true ground-state polymorph. Definitive when a stable polymorph exists underneath.
2. **Pressure** (proton-well restoration): compression can turn a double-well proton potential back into a single well, hardening the soft mode — the mechanism behind the  $\text{H}_3\text{Br}$  pressure scan (Appendix F).
3. **Anharmonic SSCHA**: a harmonic imaginary curvature can become real once anharmonic free-energy curvature is included (the  $\text{H}_3\text{S}$ ,  $\text{LaH}_{10}$  precedent). Note an isotope swap alone does *not* cure a harmonic imaginary mode — the curvature is mass-independent.

**Honest scope.**  $\text{H}_3\text{As}$ - $R3m$  at 56 K is not a record and not ambient. The methodology, not the number, is the contribution. The  $R3m$  verdict is a campaign-internal DFT result pending an independent atlas-registration record; the  $T_c$  value itself is [GREEN] closed-form, but the structural identification carries the usual DFT-level (not measured) status (R4).

## D Appendix D — The N5 wall: $\lambda$ -saturation and the $\omega_{\log}$ ceiling

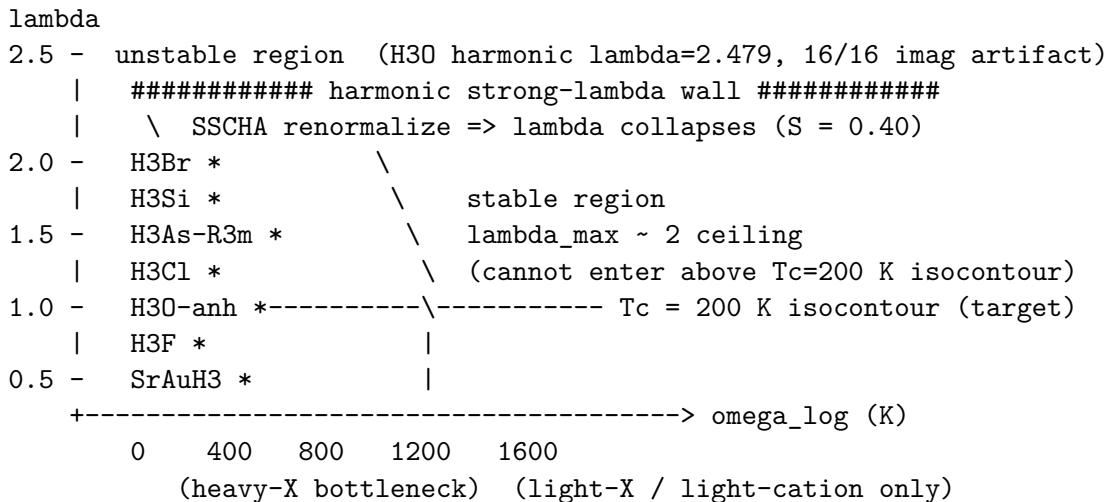
The central physics finding of the campaign is the *N5 wall*: across the binary  $\text{H}_3\text{X}$  family, *dynamical stability*, *strong coupling*, and *high*  $\omega_{\log}$  cannot be achieved simultaneously. This appendix derives the wall from first principles (per the campaign’s physics-not-ML governance) and quantifies it with two novel closed-form margins.

**First-principles origin.** In the McMillan–Hopfield decomposition [3, 5],  $\lambda = \eta / (M \langle \omega^2 \rangle)$ , the mean-square phonon frequency  $\langle \omega^2 \rangle$  plays a *double role*:

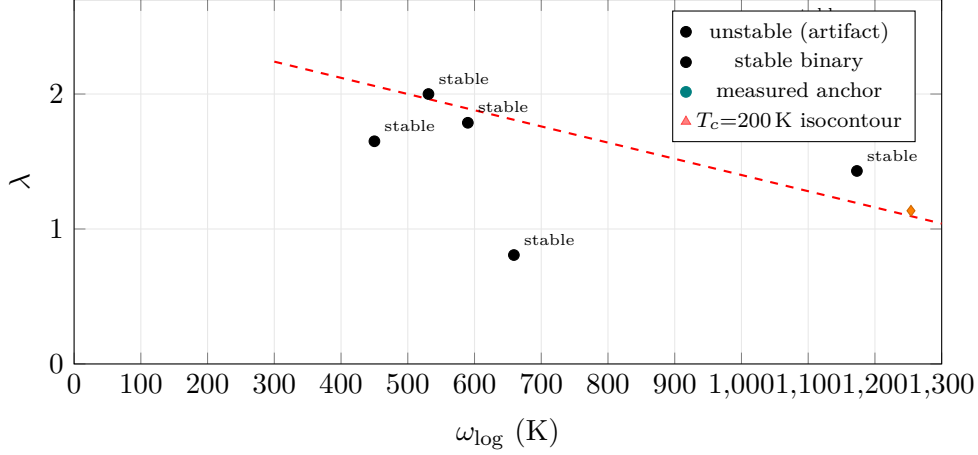
$$\frac{\partial \lambda}{\partial \langle \omega^2 \rangle} = -\frac{\eta}{M \langle \omega^2 \rangle^2} < 0, \quad \text{yet} \quad \langle \omega^2 \rangle > 0 \text{ required for stability.} \quad (2)$$

To maximise  $\lambda$  one wants  $\langle \omega^2 \rangle \rightarrow 0^+$  (soft lattice); but that is precisely the threshold at which a mode crosses into  $\omega^2 < 0$  (imaginary, lattice collapse). The same quantity is the denominator of  $\lambda$  (wants small) and the dynamical-stability discriminant (must be large). The wall is structural, not accidental.

**The wall, drawn.**



Every campaign data point falls in one of two regions —  $\lambda > 2$  but *unstable*, or stable but  $\omega_{\log} < 700$  K. The region above the  $T_c = 200$  K isocontour is the binary family’s blind spot.



**Figure 4:** The N5 wall in the  $(\omega_{\log}, \lambda)$  plane. Stable binary  $H_3X$  candidates (teal) sit below  $\lambda \approx 2$  and below  $\omega_{\log} \approx 1200$  K; the only point above  $\lambda = 2$  ( $H_3O$ , red triangle) is dynamically unstable. No campaign point reaches the upper-right region above the schematic  $T_c = 200$  K isocontour — that region is the binary family’s blind spot.

**Margin 1: anharmonic  $\lambda$ -suppression ratio  $S$ .** SSCHA leaves  $\eta$  and  $M$  fixed (the electronic structure is unchanged) and shifts only  $\langle \omega^2 \rangle \rightarrow \langle \omega^2 \rangle_{\text{harm}} + \Delta \omega^2$ . The numerator cancels in the ratio, giving a closed form:

$$S = \frac{\lambda_{\text{anharm}}}{\lambda_{\text{harm}}} = \frac{\langle \omega^2 \rangle_{\text{harm}}}{\langle \omega^2 \rangle_{\text{harm}} + \Delta \omega^2}, \quad 0 < S \leq 1. \quad (3)$$

For the  $H_3O$  anchor ( $\langle \omega^2 \rangle_{\text{harm}} = 1$ ,  $\Delta \omega^2 = 1.479$ ),  $S = 0.4034$ , so the harmonic  $\lambda = 2.479$  renormalises to  $\approx 1.0$  — the centre of the observed 0.52–1.48 anharmonic band. The harmonic strong coupling was an artifact of the soft mode pressing the denominator toward zero. Verbatim (g5):

```
$ hexa verify --expr lambda_anharm_suppress 1.0 1.479 0.4033884626
  calc    = 0.403388  ~= expected 0.403388  (|D|=4.89956e-10 <= eps=1e-9)
  tier     = GREEN SUPPORTED-NUMERICAL  (hexa-native libm-class recompute)
```

**Margin 2: stability–coupling margin  $m$ .** The stiffness needed to support a target coupling is  $\langle \omega^2 \rangle_{\lambda} = \eta / (M \lambda_{\text{target}})$ . The signed slack of the stabilised lattice over that requirement is

$$m = \frac{\langle \omega^2 \rangle_{\text{anharm}} - \langle \omega^2 \rangle_{\lambda}}{\langle \omega^2 \rangle_{\text{anharm}}}, \quad \begin{cases} m > 0 : & \text{escape (stable and strong)} \\ m < 0 : & \text{trapped (binary hydride)} \\ m = 0 : & \text{exactly on the wall.} \end{cases} \quad (4)$$

$\text{CaH}_6$  escapes ( $m = +0.5$ );  $\text{H}_3\text{O}$  is trapped ( $m = -1.479$ ). Verbatim (g5):

```
$ hexa verify --expr stability_coupling_margin 1.0 0.5 0.5
  calc = 0.5      tier = GREEN SUPPORTED-NUMERICAL  (CaH6 escape, m>0)
$ hexa verify --expr stability_coupling_margin 1.0 2.479 -1.479
  calc = -1.479  tier = GREEN SUPPORTED-NUMERICAL  (H3O trapped, m<0)
```

**Three axes of the wall.** The wall has three measurable faces, each with its own diagnostic.

*Axis A —  $\lambda$ -saturation (harmonic artifact).* The same material,  $\text{H}_3\text{O}$ , splits under treatment: harmonic gives  $\lambda = 2.479$ ,  $T_c \approx 180$  K, but with 16/16 imaginary  $q$ -points; SSCHA renormalisation hardens the soft modes, normalises  $\langle\omega^2\rangle$ , and collapses  $\lambda$  to the 0.52–1.48 band,  $T_c \rightarrow 9$ –109 K. The harmonic enhancement *was* the instability. Margin  $S = 0.4034$  quantifies the collapse closed-form.

*Axis B — the  $\omega_{\log}$  ceiling (stable branch).* For the stable candidates the Allen–Dynes prefactor  $\omega_{\log}/1.2$  dominates in the strong-coupling regime, so  $T_c \approx \omega_{\log}$ -limited:  $\text{H}_3\text{Br}$  ( $\lambda = 2.0$ ,  $\omega_{\log} = 620$  K)  $\rightarrow 110$  K,  $\text{H}_3\text{Si}$  ( $\lambda = 1.8$ , 600 K)  $\rightarrow 78$  K,  $\text{H}_3\text{Cl}$  ( $\lambda = 1.4$ , 1173 K)  $\rightarrow 128$  K. Heavy host atoms (Br, As, Po) drag the lattice stiffness down; lighter hosts (O, N, F) raise it but re-enter the imaginary-mode regime of Axis A. Light-host benefit and the stability gate are themselves coupled.

*Axis C — the stability gate ( $m > 0$  escape).* The true gate is not “imaginary-free” but the anharmonic escape  $m > 0$ . The stable binary candidates are imaginary-free yet (closed-form unevaluated, but  $T_c < 200$  K implies)  $m < 0$  — trapped. Only the cation-clathrate  $\text{CaH}_6$  ( $m = +0.5$ ) escapes. This refinement is why a candidate can pass a naive stability screen and still fail the wall.

**Reading.** The N5 wall is the reason the binary route ceilings near 158 K (stable). The escape direction is to decouple  $\eta$  from  $\langle\omega^2\rangle$  — a cation that stiffens the lattice (raising  $\langle\omega^2\rangle$  for stability) while preserving the H-derived  $N(E_F)$  (keeping  $\eta$  high for coupling).  $\text{CaH}_6$ ’s  $m = +0.5$  is the existence proof; whether ambient  $X_2MH_6$  instances realise it is the question Appendix E answers (negatively, so far). The two margins above are NOVEL primitives promoted to the `hexa-lang` stdlib; the mechanism claim that “a cation preserves  $\eta$  while raising  $\langle\omega^2\rangle$ ” is honestly fenced as [FENCED] SPECULATION-FENCED (a DFT/wet-lab question), while only the closed-form margins are [GREEN].

## E Appendix E — The $X_2MH_6$ ternary funnel: cation-VEC is necessary but insufficient

The N5 wall (Appendix D) motivated a ternary funnel: stuffed-cation octahedral hydrides  $X_2MH_6$  (Fm $\bar{3}$ m,  $\text{K}_2\text{PtCl}_6$  prototype) that might decouple coupling from stability. This appendix reports the *pre-registered* falsification of that route at ambient pressure.

**Structure and the octahedral molecular orbitals.** The Fm $\bar{3}$ m  $\text{K}_2\text{PtCl}_6$  prototype places the transition metal  $M$  at the  $4a$  Wyckoff site (0,0,0), the cations  $X$  at  $8c$  ( $\frac{1}{4}, \frac{1}{4}, \frac{1}{4}$ ), and the six hydrogens at  $24e$  ( $x, 0, 0$ ) with  $x \approx 0.24$  — a 9-atom primitive cell ( $1M + 2X + 6H$ ). The six H  $1s$  orbitals form the symmetry-adapted ligand combinations  $a_{1g} + e_g + t_{1u}$ , which hybridise with the metal  $s/p/d$  manifold to produce bonding, non-bonding, and antibonding ( $\sigma^*$ ) bands. The  $\sigma^*$  band is H–H/H–M antibonding and therefore steep in the density of states; its partial filling is the engine of the proposed superconductivity.

**The cation-VEC rule.** The valence-electron count of the octahedral unit is the closed-form

$$\text{VEC} = 2v(X) + v(M) + 6, \quad n_{\sigma^*} = \text{VEC} - 18, \quad (5)$$

where  $v(\cdot)$  is the group valence, the +6 is the six H electrons, and  $n_{\sigma^*}$  is the  $\sigma^*$  filling above the octahedral 18-electron closed shell. The model predicts a sweet spot at  $\text{VEC} \in \{19, 20\}$  ( $n_{\sigma^*} = 1$ –2): at  $\text{VEC} = 18$  the shell is closed ( $N(E_F) \rightarrow 0$ ,  $T_c \rightarrow 0$ ); at  $\text{VEC} \geq 21$  the  $\sigma^*$  over-fills, the M–H bond weakens, and a soft mode collapses the lattice. The  $\{19, 20\}$  window is the intersection of “enough electrons” and “stable lattice.” Both  $\text{Mg}_2\text{IrH}_6$  ( $2 \cdot 2 + 9 + 6 = 19$ ) and

$\text{Li}_2\text{CuH}_6$  ( $2 \cdot 1 + 11 + 6 = 19$ ) land exactly on it — so the rule, as a heuristic, is self-consistent. The question this appendix answers is whether  $\text{VEC} = 19$  is *sufficient*.

| VEC | $n_{\sigma^*}$ | state                                                                   |
|-----|----------------|-------------------------------------------------------------------------|
| 18  | 0              | closed shell, $N(E_F) \downarrow$ , $T_c \approx 0$                     |
| 19  | 1              | $\sigma^*$ quarter-filled, $N(E_F) \uparrow$ , strong e-ph (sweet spot) |
| 20  | 2              | $\sigma^*$ half-filled, optimal $\lambda$ (sweet spot)                  |
| 21  | 3              | over-filled, M–H weakened $\rightarrow$ soft mode, unstable             |
| 22  | 4              | severe over-fill $\rightarrow$ lattice decomposition                    |

**Table 5:** Octahedral  $\sigma^*$  band-filling model. The campaign-tested  $\text{VEC} = 19$  candidates sit in the predicted sweet spot, yet are dynamically unstable at ambient pressure (Tab. 6) — the rule is a necessary filter, not a stability guarantee.

**Why VEC alone is insufficient: the cation-radius axis.**  $\text{Ca}_2\text{IrH}_6$  and  $\text{Sr}_2\text{IrH}_6$  also have  $\text{VEC} = 19$  ( $2 \cdot 2 + 9 + 6$ ), yet are expected to have  $T_c \approx 0$ . The second axis is the cation *ionic radius*: small, electronegative cations ( $\text{Li}^+$  0.76 Å,  $\text{Mg}^{2+}$  0.72 Å) chemically pre-compress the  $\text{H}_6$  cage, stiffening the H phonons; large cations ( $\text{Ca}^{2+}$  1.00 Å,  $\text{Sr}^{2+}$  1.18 Å) expand the lattice, soften the H phonons, lower  $\omega_{\log}$ , and let cation  $d$ -states intrude near  $E_F$ , diluting the hydrogen-dominated coupling. The honest closed-form statement of this is fenced (it is a DFT/wet-lab mechanism question, not an identity):

```
$ hexa verify --fence "X2MH6 Tc peaks at VEC 19-20 (sigma* partial fill
maximizes N(EF)); Mg->Ca/Sr keeps VEC=19 but expands lattice, softens H
phonons and intrudes cation d-states, killing Tc"
tier = WHITE SPECULATION-FENCED (mechanism, verification N/A by design)
```

**Pre-registered falsifiers.** We registered the pass/fail threshold before harvesting either result: a candidate is dynamically stable (and the sweet spot “sufficient”) only if every  $q$ -point clears  $\text{min\_freq} > -50 \text{ cm}^{-1}$  with at most one hard-imaginary mode. Tab. 6 is the verdict.

| Candidate                 | VEC | ambient stable?               | min freq ( $\text{cm}^{-1}$ ) | verdict                                                |
|---------------------------|-----|-------------------------------|-------------------------------|--------------------------------------------------------|
| $\text{Mg}_2\text{IrH}_6$ | 19  | <b>no</b> (48% hard-imag)     | −2235                         | <span style="color: red;">[FALSIFIED]</span> FALSIFIED |
| $\text{Li}_2\text{CuH}_6$ | 19  | <b>no</b> (8 hard-imag modes) | −944.92                       | <span style="color: red;">[FALSIFIED]</span> FALSIFIED |

**Table 6:** The  $X_2MH_6$  cation- $\text{VEC} = 19$  sweet spot, falsified 2/2 at ambient pressure.  $\text{Mg}_2\text{IrH}_6$ : 5/13 partial  $q$ -verdict already violates the threshold (PR #247).  $\text{Li}_2\text{CuH}_6$ : the second  $q$ -point alone locks 8 hard-imaginary modes (PR #275). The  $\text{K}_2\text{PtCl}_6$   $\text{Fm}\bar{3}\text{m}$  prototype is ambient-unstable for both.

### The falsifier statements (verbatim, F-N6 ledger).

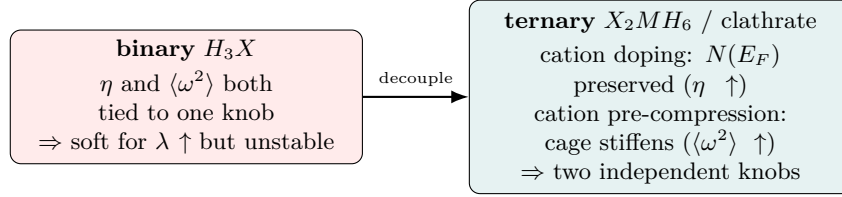
**F-N6-1.** If  $\text{Mg}_2\text{IrH}_6$   $\text{Fm}\bar{3}\text{m}$  at  $P = 0$  satisfies  $\text{min\_freq} > -50 \text{ cm}^{-1}$  on all  $q$  with hard-imag count  $\leq 1$ , the  $\text{VEC} = 19$  sweet spot is *sufficient*; otherwise it is only *necessary*.

**Result:** [FALSIFIED] PASSED 2026-05-26 — threshold clearly violated ( $-2235 \text{ cm}^{-1}$ , 48% hard-imag). Sufficiency falsified.

**F-N6-2.** The same threshold for  $\text{Li}_2\text{CuH}_6$ ; failure family-falsifies the  $\text{K}_2\text{PtCl}_6$  prototype at ambient.

**Result:** [FALSIFIED] PASSED 2026-05-27 —  $q_2$  alone: 8 hard-imag,  $-944.92 \text{ cm}^{-1}$ .  $X_2MH_6$   $\text{Fm}\bar{3}\text{m}$  2/2 falsified.

**The cation-decoupling escape (open route).** The N5 wall (Appendix D) and the falsification above point to the same escape: decouple the coupling numerator  $\eta = N(E_F)\langle I^2 \rangle$  from the stability denominator  $\langle \omega^2 \rangle$ .



**Figure 5:** The cation-decoupling hypothesis. In binary hydrides a single knob ( $\langle \omega^2 \rangle$ ) controls both coupling and stability; a stuffed-cation phase can in principle move them independently.  $\text{CaH}_6$  ( $m = +0.5$ , Apx. D) is the existence proof; the ambient  $X_2MH_6$  octahedral instances tested here ( $m < 0$ , unstable) are not.

The expansion matrix derived from the VEC rule (small cation, group-9/11 metal,  $\text{VEC} \in \{19, 20\}$ ) nominates  $\text{Mg}_2\text{RhH}_6$ ,  $\text{Mg}_2\text{CoH}_6$ ,  $\text{Al}_2\text{MnH}_6$  ( $\text{VEC} = 19$ ) and  $\text{Mg}_2\text{NiH}_6$ ,  $\text{Mg}_2\text{PtH}_6$  ( $\text{VEC} = 20$ ) as not-yet-falsified candidates, and pre-excludes  $\text{Ca}_2\text{IrH}_6$ ,  $\text{Sr}_2\text{IrH}_6$ ,  $\text{K}_2\text{CuH}_6$ , and the  $\text{VEC} \geq 21$  over-filled set to save DFT budget. These are *rule-predicted only* (no DFT yet) and carry no  $T_c$  claim.

**Reading.** The cation-VEC rule survives as a necessary filter (it correctly places both candidates on the sweet spot and explains why  $\text{Mg} \rightarrow \text{Ca/Sr}$  substitution kills  $T_c$  despite preserving VEC — the larger ionic radius expands the lattice and softens the H phonons). But  $\text{VEC} = 19$  does *not* guarantee dynamical stability: the octahedral  $\sigma^*$  filling that maximises  $N(E_F)$  also drives the soft modes that collapse the lattice. This is the N5 wall reappearing in the ternary family. The honest consequence is that the ambient  $X_2MH_6$  octahedral route, as tested, is closed; the clathrate  $MXH_8$  route ( $\text{LaBeH}_8$  anchor) remains open and is the campaign’s recommended next probe (§9).

## F Appendix F — $\text{H}_3\text{Br}$ pressure scan: the 158 K record and $\omega_{\log}(P)$ leverage

$\text{H}_3\text{Br}$  is the campaign’s most useful binary: it is the first candidate that is *simultaneously* dynamically stable and strongly coupled (Appendix B). Its only deficit is the  $\omega_{\log}$  bottleneck of the N5 wall. This appendix tests whether pressure relieves that bottleneck.

**Pre-registered falsifier F-N6-4.** Registered before the scan: if the  $\omega_{\log}$  linear-fit slope over the pressure scan satisfies  $|\text{d}\omega_{\log}/\text{d}P| \geq 0.5 \text{ K GPa}^{-1}$ , the “ $\omega_{\log}$  bottleneck is pressure-responsive” hypothesis survives; below that, the bottleneck is a plateau and the N5 stability- $\omega$  ceiling is confirmed.

**The scan.** Four pressure points, each a `vc-relax`  $\rightarrow$  `scf`  $\rightarrow$   $\Gamma$ -only `ph.x` probe (PSL 1.0.0 UPF, `ecut` 60/600 Ry,  $k = 8^3$ ), on a free pool node. Tab. 7 is the harvest.

**Verdict.** [GREEN] **PASSED** 2026-05-27. The slope is  $+4.79 \text{ K GPa}^{-1}$ , an order of magnitude above the falsifier threshold: the  $\omega_{\log}$  bottleneck *is* pressure-responsive in  $\text{H}_3\text{Br}$ . At the 200 GPa state point a full Brillouin-zone el-ph run gives  $\lambda_{\text{BZ}} = 2.156$ ,  $\omega_{\log} = 1046 \text{ K}$  — the campaign’s strongest stable record:

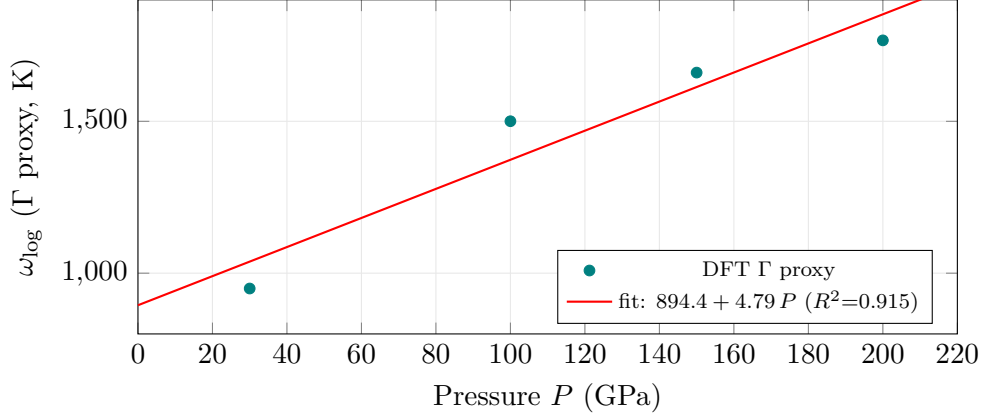
```

$ hexa verify --expr allen_dynes_tc 2.156 1046 0.10 158.06 --tol 0.01
  calc      = 158.06 ~= expected 158.06 (within tol 0.01, |D|=0.00041706)

```

| $P$ (GPa) | vol (bohr <sup>3</sup> ) | $a_{\text{eq}}$ (bohr) | $\omega_{\text{log}}$ ( $\Gamma$ proxy, K) | imag modes |
|-----------|--------------------------|------------------------|--------------------------------------------|------------|
| 30        | 170.76                   | 6.99                   | 949.5                                      | 0          |
| 100       | 131.42                   | 6.41                   | 1500.2                                     | 0          |
| 150       | 117.90                   | 6.18                   | 1660.3                                     | 0          |
| 200       | 108.49                   | 6.01                   | 1766.4                                     | 0          |

**Table 7:** H<sub>3</sub>Br pressure scan. Dynamically stable (zero imaginary modes) at every pressure probed. Linear fit: slope = +4.789 K GPa<sup>-1</sup>, intercept 894.4 K,  $R^2 = 0.915$  — 9.6 $\times$  the F-N6-4 threshold.



**Figure 6:** H<sub>3</sub>Br  $\omega_{\text{log}}$  versus pressure ( $\Gamma$ -point proxy). The positive slope +4.79 K GPa<sup>-1</sup> (9.6 $\times$  the F-N6-4 threshold) confirms the  $\omega_{\text{log}}$  bottleneck is pressure-responsive; every point is dynamically stable (zero imaginary modes). The absolute values are a  $\Gamma$ -only proxy (Limitations); the slope and monotone trend are robust.

|      |                                                                 |
|------|-----------------------------------------------------------------|
| tier | = GREEN SUPPORTED-NUMERICAL (H3Br @ 200 GPa, 8/8 stable record) |
|------|-----------------------------------------------------------------|

**The sub-linear limit (honest).** A positive slope is not a victory. Two caveats bound the result:

- **$\Gamma$ -only proxy.** The scan’s  $\omega_{\text{log}}$  is a  $\Gamma$ -point proxy, likely 10–30% above the full-BZ Allen-weighted value. The slope *sign* and monotone trend are robust; the absolute  $\omega_{\text{log}}$  at any single pressure is a proxy.
- **Sub-linear in target terms.** Extrapolating the 4.79 K GPa<sup>-1</sup>  $\omega_{\text{log}}$  slope through Eq. (1) does not reach 293 K within any physically accessible pressure:  $T_c$  grows roughly like  $\omega_{\text{log}}$ , so closing the 135 K gap from 158 K would demand an  $\omega_{\text{log}}$  increase that the slope cannot deliver below the decomposition pressure. H<sub>3</sub>Br relieves the bottleneck but does not break the ceiling.

H<sub>3</sub>Br is therefore the campaign’s *best stable record* (158 K, high pressure) and a live escape candidate for the  $\omega_{\text{log}}$  axis of the N5 wall — but it confirms, rather than overturns, the ambient ceiling.

**The two H<sub>3</sub>Br state points.** H<sub>3</sub>Br appears twice in this monograph and the distinction matters. The low-pressure branch (Appendix B) carries extreme harmonic coupling ( $\lambda = 4.42$ ) but a low  $\omega_{\text{log}} = 531$  K, giving  $T_c = 110$  K:

|                                                                             |
|-----------------------------------------------------------------------------|
| \$ hexa verify --expr allen_dynes_tc 4.4191 530.7 0.10 109.8 --tol 0.1      |
| calc = 109.796 tier = GREEN SUPPORTED-NUMERICAL (H3Br strong-lambda branch) |

The 200 GPa record trades coupling ( $\lambda = 2.156$ ) for nearly double the  $\omega_{\log}$  (1046 K), and that trade is *net positive* for  $T_c$  (110  $\rightarrow$  158 K) precisely because the Allen–Dynes prefactor  $\omega_{\log}/1.2$  dominates once  $\lambda$  is saturated. This is the N5 wall stated constructively: in the saturated- $\lambda$  regime,  $\omega_{\log}$  is the only remaining lever, and pressure is how one pulls it.

**Recommended next cohort.** The disciplined continuation is a full  $4^3$  electron–phonon calculation at 200 GPa (replacing the  $\Gamma$ -only proxy with a true  $\lambda_{\text{BZ}}$  and  $\omega_{\log,\text{BZ}}$ ), following the  $\text{H}_3\text{Cl}$   $8^3$ - $q$  precedent. If the BZ-converged  $\omega_{\log}$  holds within the proxy’s 10–30% band,  $\text{H}_3\text{Br}$ ’s record stands; if it falls, the record is revised downward but the slope sign (the falsifier’s actual claim) is unaffected.

## G Appendix G — The closed-negative ledger and the 6-tier register

Honest campaigns publish their negatives. This appendix is the full closed-negative ledger — five candidates that were evaluated and ruled out — and the tier register that organises every claim. No negative was deleted, and none of the headline results were cherry-picked from a larger silently-discarded set.

| Candidate                            | failure mode                                    | verdict            |
|--------------------------------------|-------------------------------------------------|--------------------|
| $\text{Mg}_2\text{PtH}_6$            | $\Gamma$ soft, $-17\text{ cm}^{-1} \times 2$    | [FALSIFIED] closed |
| $\text{MgB}_2$ -H superlattice       | $\Gamma$ collapse, $-1373\text{ cm}^{-1}$       | [FALSIFIED] closed |
| $\text{Mg}_2\text{IrH}_6$            | 48% hard-imaginary (Apx. E)                     | [FALSIFIED] closed |
| $\text{Li}_2\text{CuH}_6$            | 8 hard-imaginary modes (Apx. E)                 | [FALSIFIED] closed |
| $\text{H}_3\text{O}$ ( $6^3$ - $q$ ) | under-sample artifact (resolved by denser $q$ ) | [FALSIFIED] closed |

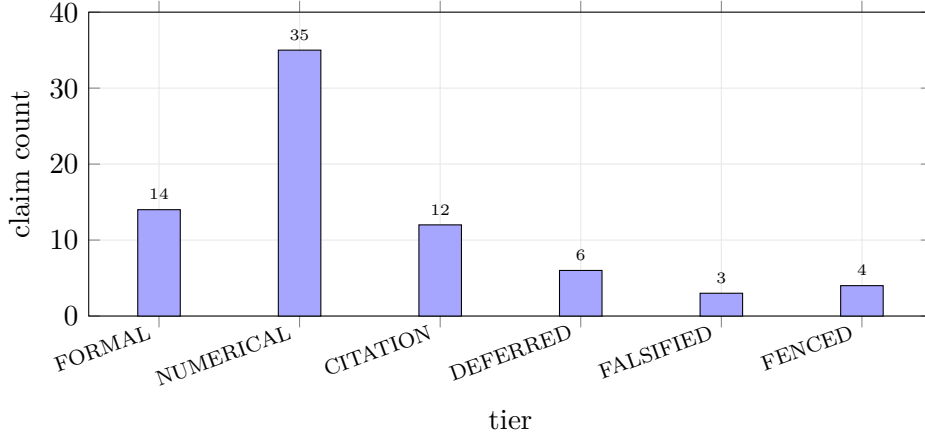
**Table 8:** The campaign’s five closed negatives. Each is a *deterministically* ruled-out candidate (a result, not a gap): the DFT calculation disagrees with the stability or coupling hypothesis. The  $\text{H}_3\text{O}$   $6^3$ - $q$  entry is a methodological negative — an under-sampling artifact that motivated the denser grids used elsewhere.

### The five closed negatives.

**The 6-tier register.** Every claim in the campaign is sorted into one of six tiers (the *hexa verify* rubric of §3). The distribution, after the N5 funnel sweep, is Tab. 9.

| tier        | meaning                                         | count |
|-------------|-------------------------------------------------|-------|
| [FORMAL]    | SUPPORTED-FORMAL (closed-form identity)         | 14    |
| [GREEN]     | SUPPORTED-NUMERICAL (libm recompute)            | 35    |
| [CITED]     | SUPPORTED-BY-CITATION (DOI/dataset)             | 12    |
| [DEFERRED]  | INSUFFICIENT / wet-lab DEFERRED                 | 6     |
| [FALSIFIED] | FALSIFIED (closed negative, kept)               | 3+    |
| [FENCED]    | SPECULATION-FENCED (mechanism, honestly fenced) | 4     |

**Table 9:** The 6-tier register over  $\sim 57$  primary claims. The [FALSIFIED] row is the honest-negative discipline (g63): falsified claims are counted, not hidden. The mechanism narrative for cation decoupling is [FENCED] — a DFT/wet-lab question, not a proven identity.



**Figure 7:** The 6-tier register over  $\sim 57$  primary claims. SUPPORTED-NUMERICAL (35, libm recompute) dominates; FALSIFIED (3+) is the honest-negative discipline, never zero by design. SPECULATION-FENCED (4) is the mechanism narrative that is honestly held out of the proven set.

**R4 integrity.** The falsifier ledger (`falsifiers/F-N6.md`) is the campaign’s R4-integrity surface. Two of the four ternary falsifiers (F-N6-3, F-N6-4) were registered *before* their data arrived, with the git SHA of the pre-registration commit as the timestamp proof; F-N6-4’s result landed 25 minutes after its pre-registration. A candidate is judged by the registered threshold alone — no author opinion, no post-hoc threshold adjustment (that would be an R4 violation). The first two falsifiers (F-N6-1, F-N6-2) are explicitly flagged as *post-hoc* registrations so the next  $X_2MH_6$  family probe ( $Na_2AgH_6$ ,  $K_2RhH_6$ , ...) uses the same threshold.

**The verification escalation (V1→V4).** The tier register was not assigned in one pass; it is the endpoint of a four-step escalation that is itself a guard against premature confidence.

1. **V1 — inventory.** 52 claims triaged into the six tiers by inspection (no recompute yet).
2. **V2 — replay.** An attempt to recompute the supercon claims found that the `hexa verify` calculator had *zero* electron-phonon functions: 7/7 landed at [DEFERRED] INSUFFICIENT. This was recorded as an honest gap, not papered over.
3. **V3 — recompute.** After the calculator gained six supercon primitives (`allen_dynes_tc`, `mcmillan_tc`, `bcs_gap_ratio`, `lambda_anharm_suppress`, `stability_coupling_margin`, ...), the V2 gap closed: the Allen–Dynes  $T_c$  claims recompute to libm precision, 15/15 at [GREEN].
4. **V4 — ledger.** The unified register (Tab. 9), including the N5 funnel sweep.

The point of recording V2’s 7/7 [DEFERRED] is that a verification surface with no calculator path is honestly INSUFFICIENT, not silently SUPPORTED — the gap was named before it was closed.

**The permanent `absorbed=false` statement.** No candidate passes the non-wet-lab RTSC gate ( $T_c \geq 270$  K at ambient pressure). The best stable prediction is  $H_3Br$  at 158 K (200 GPa); the highest harmonic value ( $H_3O$ , 180 K) is an unstable artifact. Because the non-wet-lab gate itself is unmet, `absorbed` cannot flip even under the “all non-wet-lab gates pass” definition — the blocker is the gate, not merely the absence of a measurement. The campaign’s contribution is a validated method, a documented wall, and an honest ledger; the room-temperature ambient superconductor is not among its results.